Physics 468/668: Statistical Physics **Spring 2019**

Lecture 14: Interacting systems: Cluster expansion

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

14.1 Cluster expansion

So far, we considered only systems composed of *noninteracting* entities. The results apply to only a few systems. For a contact between the theory and experiment, one must take into account the interactions between particles. We briefly talked about the possibility of including interactions when we introduced the *virial theorem*. In this chapter, we will consider to derive a general approach to handle such systems by the forms of series expansions, which was developed by Mayer.

14.2 Cluster expansion for a classical gas

Consider a monoatomic gas, its Hamiltonian is given by

$$H = \sum_i \left(\frac{p_i^2}{2m}\right) + \sum^{\text{all pairs}} u_{ij} \tag{14.1}$$

The summation in the 2nd part goes over all $N(N-1)/2$ pairs of N particles. In general, the potential u_{ij} is function of the relative position vectors r_{ij}.

The partition function is thus given by

$$\begin{aligned} Q_N(V,T) &= \frac{1}{N!h^{3N}} \int \exp\left\{ -\beta \sum_i \left(\frac{p_i^2}{2m}\right) - \beta \sum u_{ij} \right\} d^{3N}p d^{3N}r \\ &= \frac{1}{N!\lambda^{3N}} \int \exp\left[-\beta \sum u_{ij} \right] d^{3N}r \\ &= \frac{1}{N!\lambda^{3N}} Z_N(V,T) \end{aligned} \tag{14.2}$$

where $\lambda = h/(2\pi mkT)^{1/2}$ is the mean thermal length of the particles due to its kinetic energy, while $Z_N(V,T)$ is

$$Z_N(V,T) = \int \exp\left[-\beta \sum u_{ij} \right] d^{3N}r = \int \prod_{i<j} \exp(-\beta u_{ij}) d^{3N}r \tag{14.3}$$

Clearly, $Z_N(V,T)$ is related to the configurational integral of the system. If it is ideal gas,

$$Z_N(V,T) = V^N \tag{14.4}$$

To treat the nonideal case, Mayer introduced the two particle function f_{ij},

$$f_{ij} = \exp(-\beta u_{ij}) - 1 \tag{14.5}$$

Since u is a function of r_{ij}, we expect that u will have large repulsion within short distance, and go to zero at long range. So f will be -1 with in short distance, and 0 at long range.

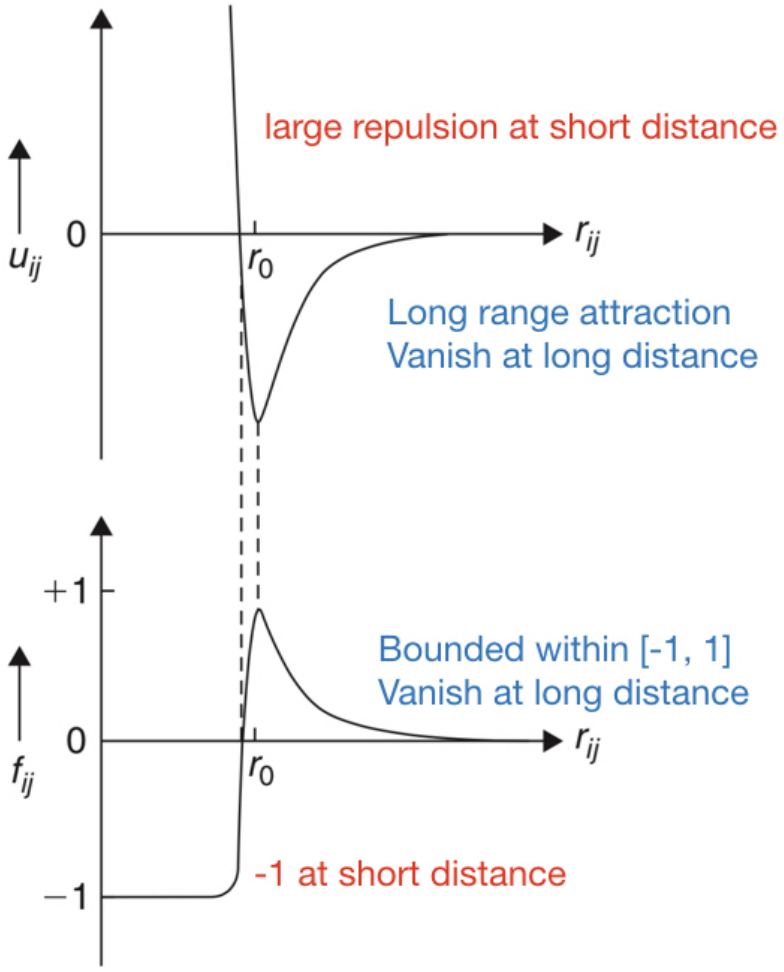


Figure 14.1: A typical plot of the two-body potential function u_{ij} and the corresponding Mayer function f_{ij}.

Now, to evaluate the integral, we expand the integrand in ascending powers of the functions f_{ij}:

$$\begin{aligned} Z_N(V,T) &= \int \prod_{i<j} (1 + f_{ij}) d^3 r_1 \cdots d^3 r_N \\ &= \int \left[1 + \sum f_{ij} + \sum f_{ij} f_{kl} + \cdots \right] d^3 r_1 \cdots d^3 r_N \end{aligned} \tag{14.6}$$

Where you can regard the expansions represent the 2-body, 3-body, ... and N-body clusters.

N-particle graph, is a collection of N distinct circles, numbered 1,2,...,N, with a number of lines linking

some (or all) of the circles.

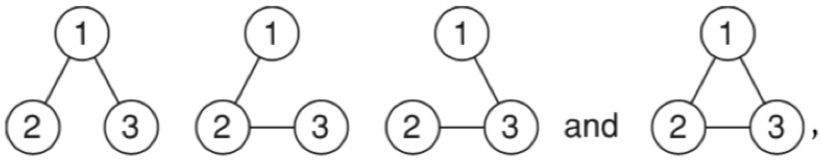
$$\int (f_\alpha f_\beta \cdots f_\lambda) d^3r_1 \cdots d^3r_N \tag{14.7}$$

A graph having the same number of linked pairs as this one but with the set $(\alpha', \beta', \cdots, \lambda')$ distinct from the set $(\alpha, \beta, \cdots, \lambda)$ will be counted as a distinct graph,

Therefore, we have

$$Z_N(V,T) = \text{sum of all distinct } N\text{-particle graphs} \tag{14.8}$$

Also, we note that some graphs have degeneracy while some not. For instance, a group of three particles leads to four 3-clusters,



of which the first three are equal. Therefore, we introduce the notation of $b_l(V,T)$

$$b_l(V,T) = \frac{1}{l!\lambda^{3(l-1)V}} \times (\text{the sum of all possible } l\text{-clusters}) \tag{14.9}$$

Some of the simple cluster integrals are:

$$\begin{aligned}
b_1 &= \frac{1}{V}\int d^3r_1 = 1 \\
b_2 &= \frac{1}{2\lambda^3 V}\iint f_{12} d^3r_1 d^3r_2 \\
&\approx \frac{1}{2\lambda^3}\int f_{12} d^3r_{12} = \frac{2\pi}{\lambda^3}\int f(r) r^2 dr \\
&= \frac{2\pi}{\lambda^3}\int (\exp(-u(r)/kT) - 1) r^2 dr \\
b_3 &= \frac{1}{6\lambda^6 V}\iiint (f_{12}f_{13} + f_{12}f_{23} + f_{13}f_{23} + f_{12}f_{13}f_{23}) d^3r_1 d^3r_2 d^3r_3 \\
&\approx \frac{1}{6\lambda^6 V}\left[3V\iint f_{12}f_{13} d^3r_{12} d^3r_{13} + V\iint f_{12}f_{13}f_{23} d^3r_{12} d^3r_{13}\right] \\
&= 2b_2^2 + \frac{1}{6\lambda^6}\iint f_{12}f_{13}f_{23} d^3r_{12} d^3r_{13}
\end{aligned} \tag{14.10}$$

Note that f_{ij} drops rapidly to zero as r_{ij} becomes large, so that, if the position of i does not happen to be within the molecular distance of the walls of the vessel,

$$\int f_{ij} dr_j^3 = \int_0^\infty f(r) 4\pi r^2 dr = \beta \tag{14.11}$$

$$b_2 = \int \beta dr_i = \beta V \tag{14.12}$$

Obviously, an N-particle graph will have a number of clusters, say m_1 are 1-clusters, m_2 are 2-clusters, and so on. The numbers $\{m_l\}$ must satisfy

$$\sum_{l=1}^{N} l m_l = N \tag{14.13}$$

However, a given set of numbers of $\{m_l\}$ does not specify a unique, single graph. it represents a collection of graphs the sum total of which may be denoted by the symbol $S\{m_l\}$. We may then write

$$Z_N(V,T) = \sum_{\{m_l\}} S\{m_l\} \tag{14.14}$$

To evaluate the sum $S\{m_l\}$. We use two observations,

- There are many different ways to assign N particles of the system to the $\sum_l m_l$ clusters
- For any given assignments, many ways to form the clusters, e.g., four ways to form a 3-cluster.

from (1), we obtain a straightforward factor of

$$\frac{N!}{(1!)^{m_l}(2!)^{m_2}\cdots} = \frac{N!}{\prod_l (l!)^{m_l}} \tag{14.15}$$

from (2), there are two steps of considerations.

1. for all l-clusters were unique, there is a power of m_l
2. for any two clusters with the same number, there should be a factor of $(m_l!)$

with the help of b_l,

$$\prod_l \left[(b_l l! \lambda^{3(l-1)} V)^{m_l} / m_l! \right] \tag{14.16}$$

Therefore, we obtain the configuration integral

$$Z_N(V,T) = N! \lambda^{3N} \sum_{\{m_l\}}{}' \left[\prod_l \left\{ \left(b_l \frac{V}{\lambda^3} \right)^{m_l} \frac{1}{m_l!} \right\} \right] \tag{14.17}$$

There is a relation

$$\prod_l (\lambda^{3l})^{m_l} = \lambda^{3\sum_l l m_l} = \lambda^{3N} \tag{14.18}$$

So we express the partition function

$$Q_N(V,T) = \sum_{\{m_l\}}{}' \left[\prod_l \left\{ \left(b_l \frac{V}{\lambda^3} \right)^{m_l} \frac{1}{m_l!} \right\} \right] \tag{14.19}$$

For the grand partition function of the system,

$$\mathcal{Q}(z, V, T) = \sum_{N=0}^{\infty} z^N Q_N(V, T) \tag{14.20}$$

Writing

$$z^N = z^{\sum_l l m_l} = \prod_l (z^l)^{m_l} \tag{14.21}$$

substituting for $Q_N(V, T)$ from Eq. 14.19

$$\begin{aligned}
\mathcal{Q}(z, V, T) &= \sum_{m_1, m_2, \cdots = 0}^{\infty} \left[\prod_{l=1}^{\infty} \left\{ \left(b_l z^l \frac{V}{\lambda^3} \right)^{m_l} \frac{1}{m_l!} \right\} \right] \\
&= \prod_{l=1}^{\infty} \left[\sum_{m_l=0}^{\infty} \left\{ \left(b_l z^l \frac{V}{\lambda^3} \right) \frac{1}{m_l!} \right\} \right] \\
&= \prod_{l=1}^{\infty} \left[\exp\left(b_l z^l \frac{V}{\lambda^3} \right) \right] \\
&= \exp\left[\sum_{l=1}^{\infty} b_l z^l \frac{V}{\lambda^3} \right]
\end{aligned} \tag{14.22}$$

and hence,

$$\frac{1}{V} \ln \mathcal{Q} = \frac{1}{\lambda^3} \sum_{l=1}^{\infty} b_l z^l \tag{14.23}$$

In the limit V goes to ∞,

$$\frac{P}{kT} = \lim_{V \to \infty} \left(\frac{1}{V} \ln \mathcal{Q} \right) = \frac{1}{\lambda^3} \sum_{l=1}^{\infty} b_l z^l \tag{14.24}$$

and

$$\frac{N}{V} = \lim_{V \to \infty} \left(\frac{z}{V} \frac{\partial \ln \mathcal{Q}}{\partial z} \right) = \frac{1}{\lambda^3} \sum_{l=1}^{\infty} l b_l z^l \tag{14.25}$$

The above two equations constitute the famous cluster expansions of the Mayer-Ursell formalism. Eliminating the fugacity z, we obtain the equation of states of the system.